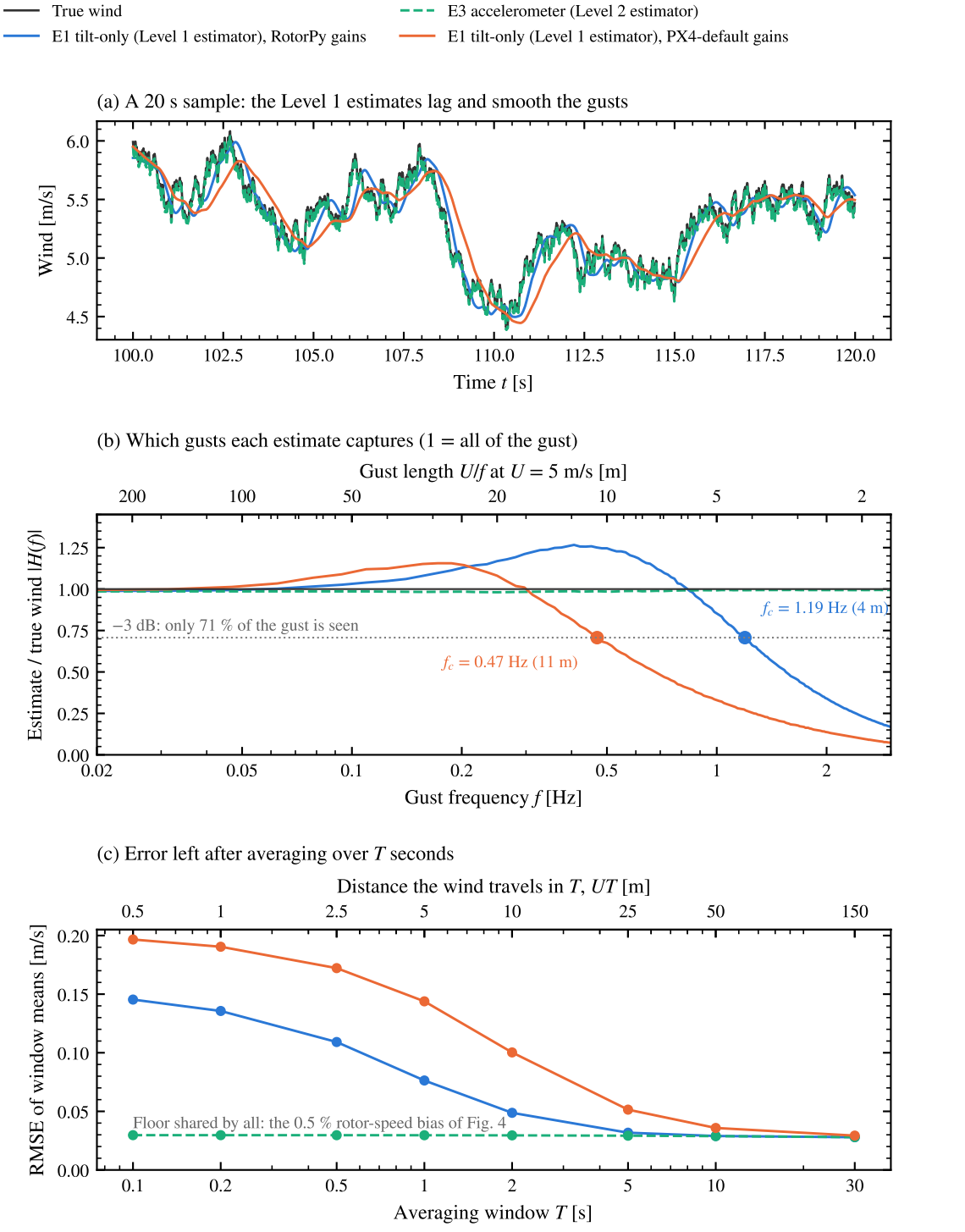
Turbulence: controller lag hides gusts shorter than 4–11 m from the Level 1 estimator; averaging over 5 s or more removes nearly all of that error



Level 2 simulation (RotorPy), Hattenberger (measured) hovering. Dryden turbulence (MIL-F-8785C, h = 10 m, W20 = U): U = 5 m/s, σ_u = 0.94 m/s, L_u = 67 m; 600 s per run, first 20 s discarded, seed 0. No sensor noise. |H| from Welch spectra, 64 s segments.